**3.30.** Write the function that eliminates the static hazard(s) in the function $F = w'z + xyz' + wx'y$.

**3.31.** Write the function that eliminates the static hazard(s) in the function $F = y'z' + wz + w'x'y$.

3.30 Solution:

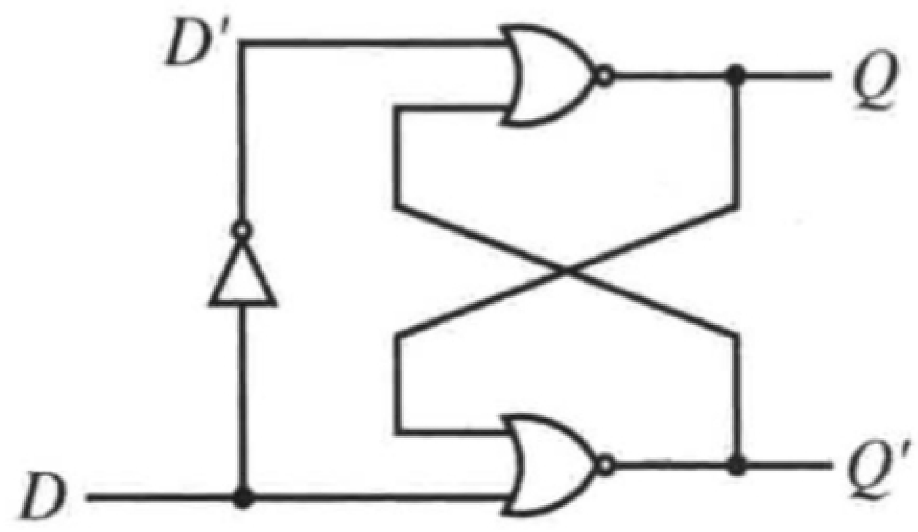
Let $A = w'z$, $B = xyz'$, $C = wx'y$

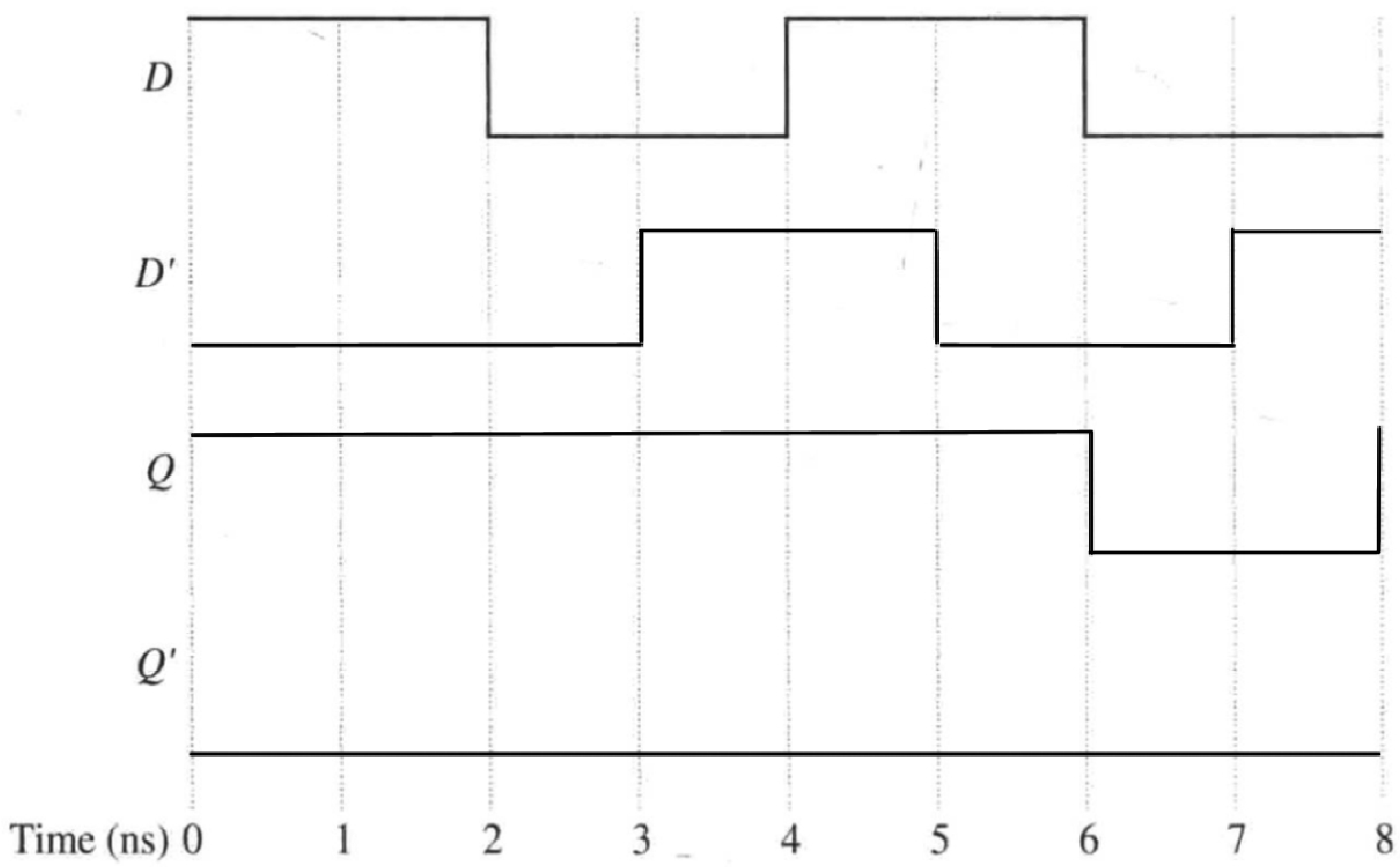
$F = w'z + xyz' + wx'y + w'xy + x'yz + wyz'$

3.31 Solution:

$F = y'z + wz + w'x'y + wy' + x'yz + w'x'z'$

5.7. Complete the following timing diagram for the following circuit. Assume that the signal delay through the NOR gates is 3 ns, and the delay through the NOT gate is 1 ns.



A benchmark program runs on a 40MHz processor. The executed program consists of 100,000 instruction, with the following instruction mix and clock cycle count. Determine the effective CPI, MIPS rate and execution time T for this program.

| Instruction type | Instruction count | Cycles per instruction |
|---|---|---|
| Integer arithmetic | 45000 | 1 |
| Data transfer | 32000 | 2 |
| Floating point | 15000 | 2 |
| Control transfer | 8000 | 2 |

$$CPI = \frac{155\,000}{100\,000} = 1.55$$

$$T = \frac{155\,000}{40 \times 10^6} = 3.875\ ms$$

$$MIPS = \frac{40 \times 10^6}{1.55 \times 10^6} \approx 25.81$$